MIL-W-23514A(SHIPS)
31 May 1967
SUPERSEDING
MIL-W-23514(SHIPS)
25 February 1963
(See 6.5)

MILITARY SPECIFICATION

WRENCH, SPANNER, ADJUSTABLE HOOK: NONSPARKING

AND NONMAGNETIC

1. SCOPE

1.1 Scope.- This specification covers one size of adjustable hook spanner wrench made of non-ferrous alloys having a low sparking hazard and a low magentic permeability.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-T-580 - Tool Steel, Carbon and Carbon-Vanadium.

MILITARY

MIL-G-5572 - Gasoline, Aviation: Grade 80/87, 91/96, 100/130, 115/145.
MIL-H-15424 - Hand Tools, Packaging of.
MIL-I-17214 - Indicator, Permeability: Low-Mu(Go - No - Go).

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5120

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3. REQUIREMENTS

3.1 Qualification.- The spanner wrenches furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Material.- The spanner wrenches shall be fabricated of a non-ferrous metal alloy which will not cause an explosion when subjected to the sparking test specified in 4.6.3 and which shall have passed the magnetic permeability test specified in 4.6.4. The alloy shall not contain more than 5 percent iron in its chemical composition and the iron shall be well diffused throughout the alloy.

3.3 Hardness.- The handle, hook and pin of the spanner wrenches shall have a minimum hardness of 90 on the Rockwell "B" scale.

3.4 Proof loads.- The spanner wrenches shall withstand the proof loads indicated below without perceptible deformation, formation of cracks, fracture of a part, or indentation or gouging of the bearing surfaces. The spanner wrenches shall be tested on the 4 inch mandrel (see figure 2) first and then on the 6-3/8 inch mandrel, under the test loads indicated for the respective size of mandrel.

Mandrel size inches	Proof load inch-pounds
4	7,500
6-3/8	10,000

3.5 Design.- Spanner wrenches shall conform to Table I and be similar to figure 1.

3.6 Marking.- Each component of the spanner wrench shall be marked in a plain and permanent manner with the name, abbreviation, or chemical symbol of each alloy used in the fabrication of this part of the wrench. Each wrench handle shall also be marked with the manufacturer's name or with a trade mark of such known character that the source of manufacture may be readily determined.

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TABLE I - Adjustable spanner wrench.

Capacity for circle diameters	Weight (max.)	Overall dimensions (hook fully opened)	Handle thickness	HOOK DIMENSIONS		
		Length		Depth	Width	Thickness
inches	pounds	inches ± 1/2	inch ± 1/8	inch ± 1/16	inch ± 1/16	inch ± 1/16
4 to 6-3/8	1-3/4	12-1/4	7/16	1/4	7/16	5/8

3.7 Finish.- All surfaces shall be finished in accordance with good commercial practice.

3.8 Workmanship.- Workmanship of the wrenches furnished shall be of the quality prevailing among manufacturers normally producing wrenches of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests.- 1/ Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the examination of 4.5.1 and the tests of 4.6.1 through 4.6.4.

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.4).

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4.2.1 Samples for qualification tests.- Two spanner wrenches shall be submitted for qualification tests.

4.3 Certification.- The supplier shall furnish a certified report of the chemical composition of the material used in the fabrication of the end item, certifying that the composition of each lot or batch of the material used is the same as that used in the product for which qualification was obtained. The Government reserves the right, at its option, to verify the correctness of such certificates.

4.4 Sampling for quality conformance inspection.-

4.4.1 Lot.- All nonsparking, nonmagnetic spanner wrenches offered for delivery at one time shall be considered a lot for the purpose of quality conformance inspection.

4.4.2 Sampling.- A random sample of nonsparking nonmagnetic spanner wrenches shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-2. The acceptable quality level (AQL) shall be 2.5 percent defective.

4.5 Quality conformance inspection.- Quality conformance inspection shall be performed on each sample wrench selected in accordance with 4.4.2. Quality conformance inspection shall consist of the following:

(a) Examination (4.5.1).
(b) Tests (4.5.2).
(c) Inspection of preparation for delivery (4.7).

4.5.1 Examination.- Each of the sample nonsparking, nonmagnetic spanner wrenches selected in accordance with 4.4.2 shall be examined to verify compliance with the requirements of this specification. Examination shall be conducted in accordance with table II. Nonconforming wrenches shall not be offered for delivery and if the number of nonconforming wrenches exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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Table II - Classification of defects.

Categories	Defects
Major:	
101	Type and capacity of spanner wrench not as specified; wrench not adjustable as specified.
102	Dimensions of spanner wrenches (overall, handle thickness, hook depth, width, and thickness) not within the specified tolerances.
103	Movable jaw not free pivoting; jaw sticks, binds, or evidences excessive looseness; hinge pin not well secured.
104	The name, abbreviation, or chemical symbol of each alloy used not permanently and plainly marked on each wrench component as specified.
Minor:	
201	Marking of manufacturer's name or trademark missing, illegible, incorrect, or not permanent.

4.5.2 Tests for quality conformance.- Quality conformance tests shall consist of the tests specified in 4.6.1, 4.6.2, and 4.6.5.

4.6 Test procedures.-

4.6.1 Hardness test.- Each of the sample wrenches selected in accordance with 4.4.2 shall be subjected to a Rockwell hardness test to determine conformance with the hardness requirements of this specification (see 3.3). Failure to comply with the minimum hardness specified shall be cause for rejection.

4.6.2 Proof load test.-

4.6.2.1 Test mandrel.- The mandrel used in load tests shall be fabricated similar to figure 2. The mandrel shall have a hardness greater than that of the wrenches under test and shall be fabricated from a material which will withstand the test loads without permanent deformation.

4.6.2.2 Application of test load.- The wrench shall be mounted on the mandrel as shown in figure 2 and the mandrel fixed between two pipe vises. The load shall be applied to the wrench handle, as far away from the hook opening as practicable (see 3.4 for proof loads). Load may be applied by dead weights or a suitable torque testing machine. Failure of the spanner wrenches to withstand the specified proof loads without perceptible deformation, formation of cracks,

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fracture of any part, or indentation or gouging of the bearing surfaces shall be cause for rejection.

4.6.3 Sparking test.-

4.6.3.1 Apparatus.- The sparking test shall be conducted in an explosion chamber having a volume of 4.5 cubic feet. The chamber shall be furnished with the following: a hot plate for vaporizing gasoline: a means for admitting gasoline and oxygen at atmospheric pressure: a fan for mixing the gasoline vapors and oxygen: two electrodes or other suitable means for providing an electric spark: a 3-inch diameter by 1/4-inch thick metal wheel having a coarsely knurled surface on the circumference and rotating at a speed of 10,000 revolutions per minute: and a fixture for holding the test specimen against the circumference of the metal wheel while applying pressure from the outside of the chamber. The metal wheel shall be made from carbon tool steel conforming to class WI-09 of QQ-T-580 and heat treated to a Rockwell hardness of C-60 to 65. The gasoline to be used during these tests shall be aviation gasoline conforming to grade 100/130 of MIL-G-5572. An oxygen analyser shall be used for determining the percentage of oxygen in the chamber.

4.6.3.2 Procedure.- If practicable, a test specimen, approximately 1/4-inch in diameter by 2 inches long, shall be made from material used in the sample tool and shall be subjected to the sparking test. If the tool contains a previously untested, questionable material in a part too small to provide a full sized test specimen or a specimen which will provide the desired test conditions, the laboratory may request the supplier to furnish a sample of the material which is large enough to provide the required test specimen. The supplier shall certify in writing that the sample material is identical to the material used in the part of the sample tool which was too small for testing. With the specimen mounted in the holder fixture and chamber sealed, oxygen shall be admitted into the chamber until the oxygen content in the air reaches 50 percent. A pan resting on the hot plate shall then be filled with 20 cubic centimeters of gasoline through an opening at the top of the chamber, the opening sealed, the fan turned on, and the gasoline evaporated. After the gasoline is completely evaporated, the fan shall be turned off and the specimen held against the rotating wheel with a force of 2.5 pounds. If no explosion occurs before 5 seconds, the mixture shall then be exploded by means of the electric spark to demonstrate the presence of an explosive mixture. Three runs shall be made with the test specimen and if the specimen explodes the gasoline and oxygen-enriched air mixture during any of these runs, the tool represented by the specimen tested shall be disapproved for use as a nonsparking tool.

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4.6.4 Magnetic permeability test.-

4.6.4.1 Apparatus.- The magnetic permeability test shall be conducted with a permeability indicator similar or equal to that covered by MIL-I-17214 according to operating instructions accompanying each indicator. The magnetic permeability shall not be above 1.2 of the indicator scale. A permeability reading in excess of 1.2 on the indicator scale shall be cause for rejection.

4.6.5 Weight test.- Each sample wrench shall be weighed to determine compliance with the maximum weight requirements of table I. Wrenches weighing more than the 1-3/4 pounds specified shall be rejected.

4.7 Inspection of preservation, packaging, packing and marking for shipment and storage.- Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with 5.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.- Preservation, packaging, and packing shall be in accordance with MIL-H-15424. Levels of preservation and packaging and level of packing shall be as specified (see 6.2).

5.2 Marking.- Marking shall be in accordance with MIL-H-15424.

6. NOTES

6.1 Intended use.- The nonsparking, nonmagnetic spanner wrenches furnished under this specification are for use in turning spline type fittings on fuel hoses, fuel lines and the like in the presence of flammable gases, dusts, or explosives, and on similar operations on mine sweepers where low magnetic permeability is required.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation and packaging and level of packing required (see 5.1).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 23514, whether or not such products have actually been so listed by that date. The attention of the

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suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application of Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4).

6.4 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy-SH
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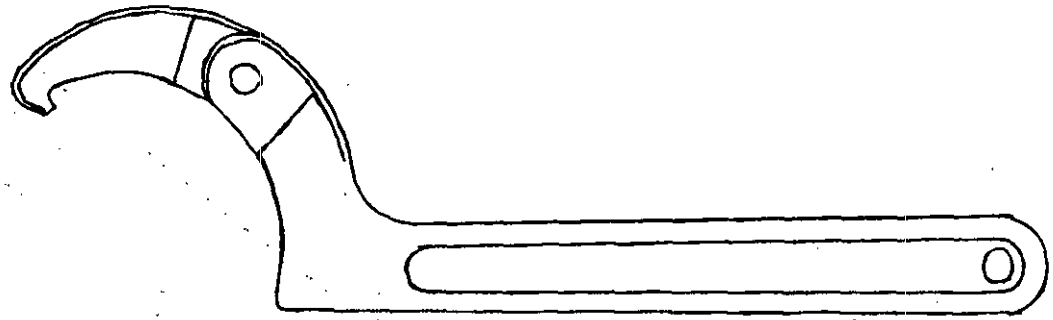


Figure 1 - Adjustable Spanner Wrench

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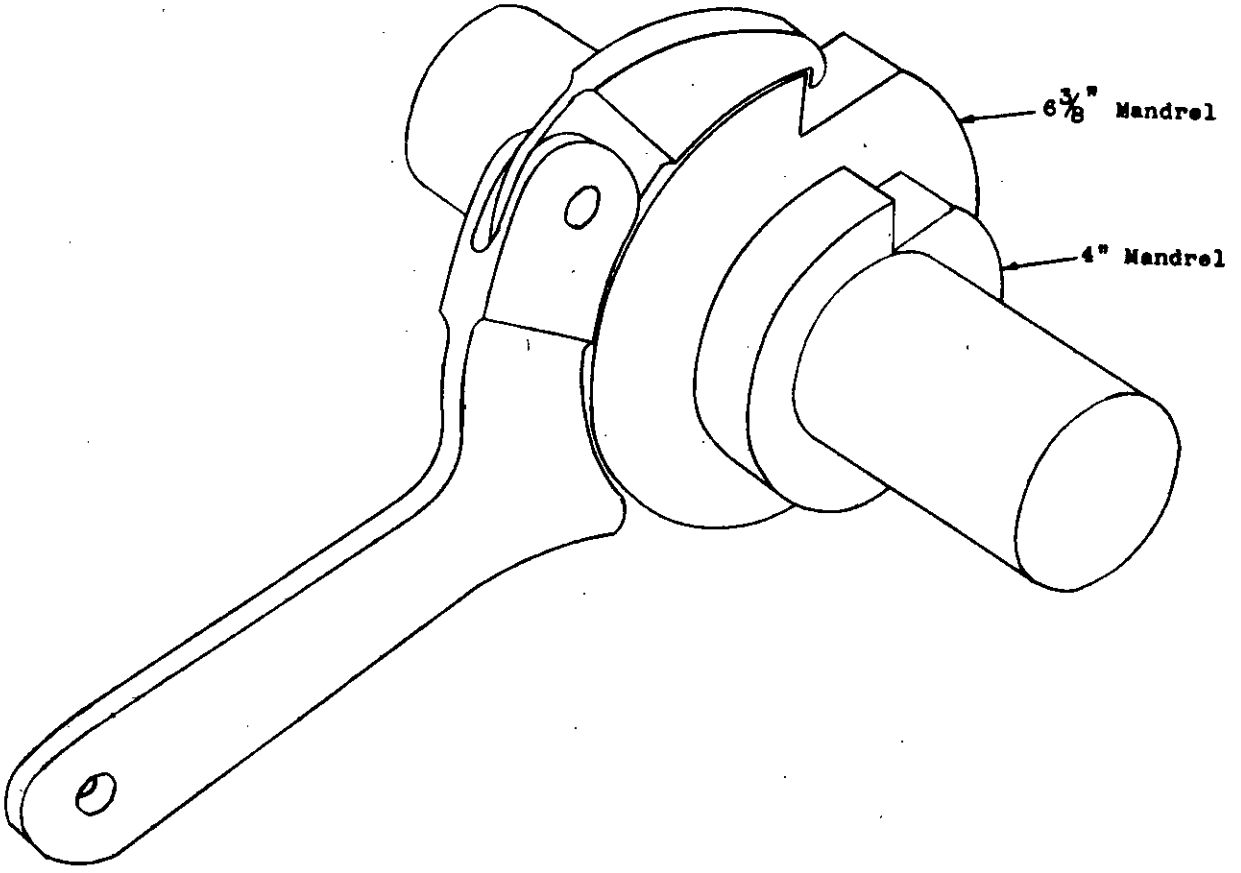


Figure 2 - Mandrel used for bending moment determinations on adjustable hook spanner wrenches

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